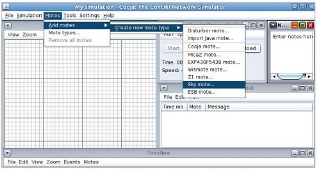

Figure 6: Invoking wireless and RFID motes in the Cooja simulator

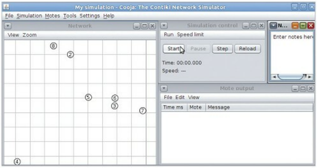

Figure 9: Viewing wireless motes with positions in Cooja

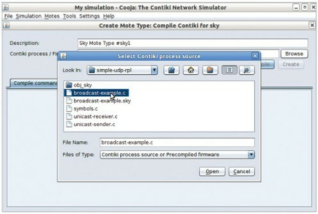

Figure 7: Importing C source code to recompile in Cooja

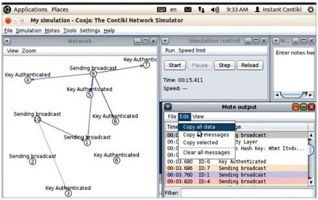

Figure 10: Running simulation and behaviour analytics of motes

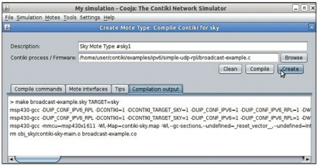

Figure 8: Compiling C source code for the desired behaviour of wireless motes in Cooja

set, which include the name of the simulation, radio medium, startup delay and random seed.

Once the basic layout and working environment is ready, you need to import the RFID tags, sensor nodes or any other wireless devices that are to be connected and have to communicate over the IoT. In wireless networking and IoT, these are known as motes. There are many types of motes in Cooja that can be programmed.

Physical motes, too, can be connected by using ports on the system so that real-time interfacing can be done. Every mote, with the base properties and programming APIs, is specified in the C source code at the back-end of Cooja. These C source code files can be customised and recompiled to get the new or desired output from these motes.

After compiling C code, a number of virtual motes can be imported in the simulation area so that the transmission of radio signals can be viewed and analysed.

In this simulation of the IoT network, the scenario of a dynamic key exchange between the motes is done. Here, the dynamic security key is generated and authenticated for communication. In the IoT, for reasons of security, it is necessary to devise and implement the protocols and algorithms by which the overall privacy and security in communication can be enforced to avoid any intrusions. As the IoT can be used for military applications, it becomes mandatory to work on highly secured key exchange algorithms with the dynamic cryptography of security keys.

Once the simulation is complete, the network log files, which include the source and destination motes, the time, and the overall activities performed during simulation are analysed. In the Mote Output Window, the log data can be copied and further analysed using data mining and machine learning tools for predictive analytics. END

By: Dr Gaurav Kumar

The author is the MD of Magma Research and Consultancy Pvt Ltd, Ambala. He is associated with various academic and research institutes, where he delivers lectures and conducts technical workshops on the latest technologies and tools. You can contact him at *kumargaurav.in@gmail.com*. Website: *www.gauravkumarindia.com*.